

QB-TimeWarp Templates Structure

Last Updated: 2026-05-20
Status:  Corrected - Templates as Golden Copy

Directory Hierarchy (2 Levels)

QB-TimeWarp uses a **2-level self-contained file structure** to protect certified templates and enable clean, repeatable testing:

```
C:\QB-TimeWarp\
├── Templates\           ← Level 1: GOLDEN COPY (READ-ONLY)
│   ├── QB2021_Blank\
│   │   └── Blank_Template.qbw (13.25 MB)
│   └── README.txt       ← Explains golden copy structure
└── Working\            ← Level 2: OPERATIONAL (MODIFIED BY TESTS)
    ├── Source\
    │   └── [Customer .qbw file - copied in]
    └── Target\
        └── [Blank template - refreshed from Templates\]
```

Level 1: Templates\ Directory (GOLDEN COPY)

Location: C:\QB-TimeWarp\Templates\
Purpose: Contains certified, pristine QuickBooks blank templates
Status: READ-ONLY - NEVER modified by QB-TimeWarp automation
Source: Created manually via QuickBooks GUI by Joseph
Created: 2026-05-20
Usage: Source for `--refresh` operations only

What It Contains

Path	File	Size	Purpose
Templates\QB2021_Blank\	Blank_Template.qbw	13.25 MB	Certified blank QB 2021 template
Templates\	README.txt	2.6 KB	Directory documentation

Critical Rules

-  **DO:** Use `--refresh` flag to copy FROM Templates\ to Working\
-  **DO:** Keep Templates\ as pristine master copy

- **✗ DON'T:** Run QB-TimeWarp.exe pointing directly at Templates\ files
- **✗ DON'T:** Modify Templates\ files manually
- **✗ DON'T:** Copy files from Desktop into Templates\ (Desktop is obsolete)

Level 2: Working\ Directory (OPERATIONAL)

Location: C:\QB-TimeWarp\Working\

Purpose: Active working copies for all migration operations

Status: OPERATIONAL - Gets refreshed from Templates\

Usage: All export/import/validation happens here

Safety: Can be safely overwritten, corrupted, or tested

What It Contains

Subdirectory	Content	Source
Working\Source\	Customer QB 2023 file	Copied from customer-provided file
Working\Target\	Blank QB 2021 template	Refreshed FROM Templates\ directory

Workflow

1. --refresh flag triggered
 - Templates\QB2021_Blank\Blank_Template.qbw
 - COPIED TO
 - Working\Target\Blank_Template.qbw
2. Customer file copied
 - Customer provides .qbw file
 - COPIED TO
 - Working\Source\Customer.qbw
3. Migration operations
 - ALL operations happen in Working\ directory
 - Templates\ remains completely untouched
4. If test fails
 - Run --refresh again
 - Fresh copy from Templates\ Working\
 - Start over with pristine template

Self-Contained System

Everything QB-TimeWarp needs is in C:\QB-TimeWarp\ except:

- Customer source .qbw files (provided by client - ONLY external input)
- QuickBooks application itself (installed separately on Windows)

What About Desktop?

Desktop was used temporarily for initial template certification only.

- ❌ Desktop is **NOT** part of the normal workflow
- ❌ Desktop files are **obsolete** after Templates\ was created
- ✅ *Templates* IS the source of truth
- ✅ All operations use Templates\ → Working\ flow



Migration Workflow (Strictly Enforced)

Step 0: Prepare Environment

- └ Run: QB-TimeWarp.exe --refresh
- └ Action: Templates\ → Working\ (fresh copies)

↓

Step 1: Load Schema (NO QB NEEDED)

- └ Load cached schema from Schemas\QB_Schema_QB_2021.json

↓

Step 2: Export from QB 2023

- └ Open: Working\Source\Customer.qbw (QB 2023)
- └ Export all data → ExportedData\
- └ CLOSE QB 2023 completely

↓

Step 3: Transform Data

- └ SDK 16.0 → 15.0 field adaptation
- └ Strip timezone offsets
- └ Apply all fixes

↓

Step 4: Import to QB 2021

- └ Open: Working\Target\Blank_Template.qbw (QB 2021)
- └ Import all data (staged by dependency order)
- └ CLOSE QB 2021 completely

↓

Step 5: Validation

- └ Verify balances, counts, integrity
- └ Generate migration report



--refresh Flag Behavior

The --refresh flag is the key to the self-contained system:

Command:

```
C:\QB-TimeWarp\QB-TimeWarp.exe --refresh
```

What It Does:

1. *Preserves Templates* (READ-ONLY - never touched)
2. *Overwrites Working\Target* with fresh copy from Templates\
3. *Preserves Working\Source* (customer file remains)
4. **Result:** Clean target template, ready for new migration test

When to Use:

-  Before every migration test run
-  After a failed test (to reset target template)
-  When target template becomes corrupted
-  When you need a guaranteed pristine starting point

When NOT to Use:

-  If you want to preserve target modifications for inspection
-  During active debugging (will lose current state)



File Size Reference

File	Location	Size	Status
Blank_Template.qbw	Templates\QB2021_Blank\	13.25 MB	 Certified blank
Joshs_Gold_Coast_II_2023.qbw	[Customer-provided]	~30 MB	 Test file
Air_Masters_QB_2023.qbw	[Customer-provided]	~360 MB	 Production (don't test!)



Safety Features

1. Templates\ Protection

- Never modified by automation
- Always available as pristine backup
- Source of truth for all tests

2. Working\ Isolation

- All destructive operations happen here
- Can be reset with single --refresh command
- Protects both Templates\ and customer originals

3. Customer File Protection

- Original customer files never modified
- Working copies created in Working\Source\
- Originals remain safe in their original location

4. No Desktop Dependency

- System doesn't rely on Desktop paths
- All paths are application-local (C:\QB-TimeWarp)
- Installer-friendly structure



Key Takeaways

Concept	Location	Status	Modification
Golden Copy	Templates\	Pristine	NEVER
Operational	Working\	Active	Always
Customer Files	External	Protected	NEVER
Desktop	Obsolete	Unused	N/A

Remember:

- Templates\ = GOLDEN COPY (never modified)
- Working\ = OPERATIONAL (gets refreshed from Templates)
- Customer files = EXTERNAL (only thing copied from outside)
- Desktop = TEMPORARY (used only for one-time certification - now obsolete)

For questions, contact Joseph (MSP Owner, 40-year programmer)